质量合格证明书
QUALITY QUALIFIED CERTIFICATE

湖北志联模具科技有限公司
Hubei Zhilian Mould Technology Co,Ltd

合同号 Contract No.		订货单位 Purchaser		产品名称 Name	工模具钢

牌号 Steel Grade	炉号 Heat No.	规格 Size	捆/支数 Number of Bundles/	重量 Weight	产品标准 Specification	交货状态 Condition of Delivery	加工用途 Purpose of Further Processing
H13电炉	5C323	610*220			GB/T1299-2014	球化退火	冷加工

化学成份(%) Chemical Composition

C	Si	Mn	P	S	Cr	Mo	V	Ni	Cu	W	Ti	Al
0.44	0.92	0.28	0.019	0. 005	4.95	1.11	0.81					

B	Co	Nb	H_2	O_2	N_2	Pb	Sn	Bi	Sb	As	Se	Re

试样热处理(℃) Sample Heat Treatment			力学性能 Mechanical Property							低倍组织(级) Macrostructure(Grade)							
正火 Normalled	淬火 Quenched	回火 Tempered	比例极限 Yield Point Rp(MPa)	屈服强度 Yield Strength Rel (MPa)	抗拉强度 Ultr-Tensile Strength Rm(MPa)	伸长率 Elongatier A(%)	面缩率 Reduction Of area Z(%)	冲击功 Impace AK(J)	硬度 Hardness HRC or HB	一般疏松 Ordinary Porosity	中心疏松 Center porosity	方形偏析 Square Segregation	点状偏析 Spot Segregation	出厂硬度 Surface Hardness (HB)	顶锻 Upsetting	超声波检验 Ultrasonic examination	断口 Fracture
											1.0	0.5		合格		GB/T4162-91	
																E/e级合格	

非金属杂物(级) Non-metallic Inclusions (Grade)										碳化物(级) Carbide (Grade)				球化(级) Spheroidization (Grade)	脱碳层 Decarburized Layer (mm)	δ相	显微孔隙(级) Microporosity (Grade)	晶粒度(级) Grainsize (Grade)
氧化物 Oxide	硫化物 Sulphuret	点状 Spot Inclusion	A		B		C		D									
			细 T	粗 H	细 T	粗 H	细 T	粗 H	细 T	粗 H	带状 Bands	液析 Liquid Segregation	网状 Network	不均匀性 non-uniformiry				
			0.5	0	1.0	0.5	0	0	0.5	0	合格	0. 5						

注：经检验与试验，本产品符合合同标准之规定

Note: Through inspection and testing, the quality of this product conforms to the stipulations of the related contract and standard(s).

判定员盖章
Stamp of Judge
签发日期
Date of Issue 2023.4.12

质量部门盖章
Stamp of Department of Quality Assurance

湖北志联模具科技有限公司 Hubei Zhilian Mould Technology Co,Ltd 质量证明书专用章